Microeconomics

Lecture 2 · The consumer problem

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From preferences to an actual choice

- Budget sets, existence and Marshallian demand.

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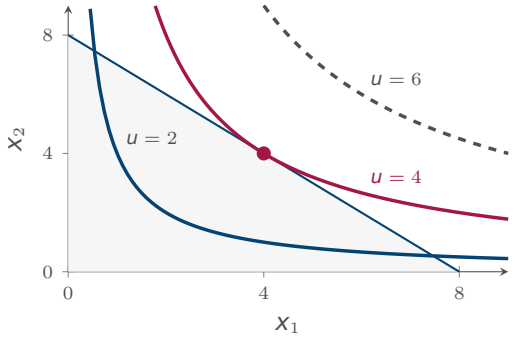
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From preferences to an actual choice

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- The Lagrangian, saddle points and Kuhn–Tucker conditions, with two proofs.
- Use the structure: budget exhaustion, demand properties, worked examples, envelopes and Roy’s identity.

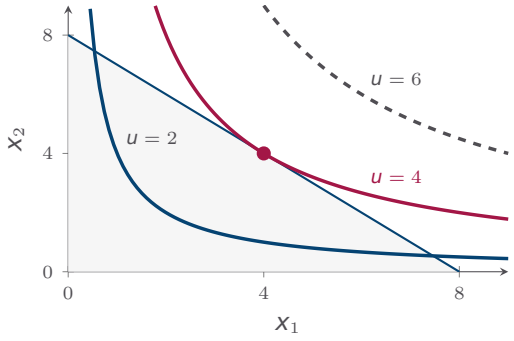
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$p = (1, 1)$, $m = 8$.

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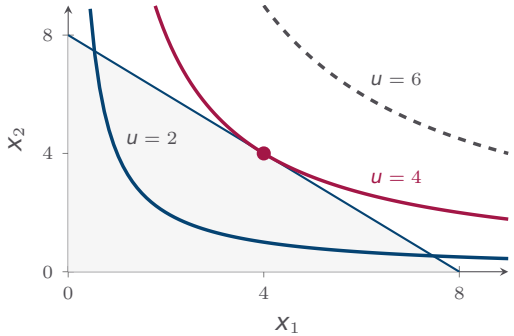


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Best bundle: $(4, 4)$.

Utility is 4.

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Here $p \gg 0$ and $m > 0$. The feasible set is

$$B(p, m) = \{x \geq 0 : p \cdot x \leq m\}.$$

What did we assume when we wrote that?

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What changes for a subscription, a quantity discount or an indivisible purchase?

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The same bundles are affordable. Only the currency units have changed.

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A closed and bounded subset of finite-dimensional Euclidean space is compact.

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Existence does not yet imply uniqueness, an interior solution or differentiability of demand.

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Can you give a case with multiple maximisers even though the budget is compact?

For $u = x_1 + x_2$, $p = (1, 1)$ and $m = 1$, every bundle on $x_1 + x_2 = 1$ is optimal.

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It is the earlier choice rule applied specifically to budget menus: $D(p, m) = C(B(p, m))$.

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Example: $(p, m) = ((2, 1), 12)$ and $((200, 100), 1200)$.

Let a multiplier enforce the budget

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One variable chooses consumption; the other enforces the budget. Their directions of optimisation differ.

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This identity does not authorise exchanging max and inf.

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Left inequality: x^* maximises ℓ over the consumption domain, holding λ^* fixed.

Right inequality: λ^* minimises ℓ over nonnegative multipliers, holding x^* fixed.

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Therefore $\lambda^*(m - p \cdot x^*) = 0$ and $\ell(x^*, \lambda^*) = u(x^*)$.

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This direction needs no concavity or differentiability: the saddle inequalities are already global.

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We now make the bounds $x_i \geq 0$ explicit using multipliers μ_i. → KKT back to a saddle

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If both goods are consumed in a smooth optimum, these marginal returns must agree.

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For the smooth argument, assume u is differentiable on an open neighbourhood of the candidate.

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Stationarity is only one group of conditions. We must also check signs, feasibility and slackness.

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Complementary slackness:

$$\lambda^*(m - p \cdot x^*) = 0, \qquad \mu_i^* x_i^* = 0.$$

Complementary slackness: read the products

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Does a binding constraint always have a strictly positive multiplier?

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This is an interior condition, not a universal rule for every optimum.

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Why would a rule that sets every marginal return equal miss some solutions?

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Quasiconcavity alone does not make every stationary point optimal. → Sufficiency proof → Counterexample

Theorem 1: KKT is necessary

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This includes corner optima and does not require concavity.

We prove this directly: construct the multipliers, prove their signs, then verify complementary slackness.

Construct the budget multiplier from an optimum

Take a nonzero optimum x^*; write $u_i = \partial u / \partial x_i$. Choose j with $x_j^* > 0$.

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Feasible spending transfers between two consumed goods show that $u_i(x^*)/p_i = \lambda^*$ whenever $x_i^* > 0$.

Also $\lambda^* \geq 0$: otherwise $u_j(x^*) < 0$, and consuming slightly less of good j would be affordable and strictly improve utility.

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The reverse strict inequality is ruled out by the reverse transfer. Thus $u_i(x^*)/p_i = u_j(x^*)/p_j = \lambda^*$.

Construct the quantity-bound multipliers

For each good, define

$$\mu_i^* := \begin{cases} 0, & x_i^* > 0, \\ \lambda^* p_i - u_i(x^*), & x_i^* = 0. \end{cases}$$

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The remaining question: why are these constructed μ_i^* nonnegative? We have not assumed their sign.

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For the consumed good j, $\lambda^* = u_j(x^*)/p_j$. Therefore

$$\frac{u_i(x^*)}{p_i} > \frac{u_j(x^*)}{p_j}.$$

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The unconsumed good would give more marginal utility per unit of spending.

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$$\frac{d}{d\varepsilon} u(x(\varepsilon))\Big|_{0+} = \frac{u_i(x^*)}{p_i} - \frac{u_j(x^*)}{p_j} = -\frac{\mu_i^*}{p_i} > 0.$$

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A sufficiently small transfer strictly improves utility: contradiction. Thus $\mu_i^* \geq 0$, with equality allowed.

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For every i, the construction gives $\mu_i^* x_i^* = 0$: either the quantity or its multiplier is zero.

Feasibility came from x^*; stationarity, multiplier signs and slackness have now all been proved.

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Choose $\lambda^* = 0$ and $\mu_i^* = -u_i(0) \geq 0$. Stationarity and both complementary-slackness products hold.

This completes the necessity proof, including all corners and the zero bundle.

Theorem 2: concave KKT is sufficient

Let u be concave on $\mathbb{R}^L_+$ and differentiable on a neighbourhood of a candidate x^*.

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The global argument uses concavity of the chosen index, not merely convexity of preferences.

Fix the multipliers and look only at quantities

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The constructed stationarity conditions give $\nabla f(x^*) = 0$. Why is that a global maximum?

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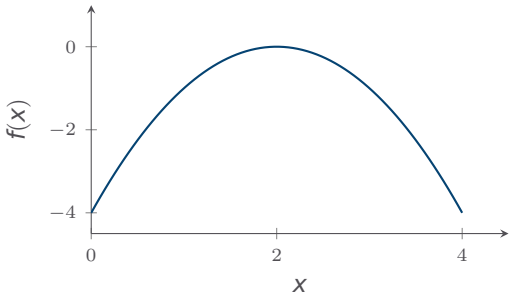
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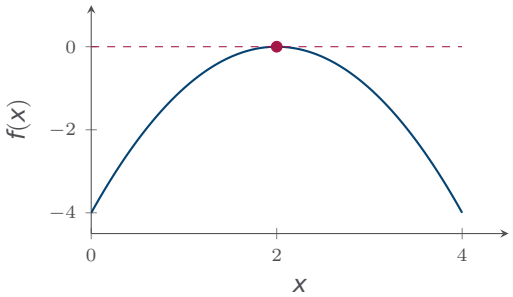
Thus x^* globally maximises $\mathcal{L}(\cdot, \lambda^*, \mu^*)$ on that domain. Uniqueness is not asserted.

A horizontal tangent above the whole graph



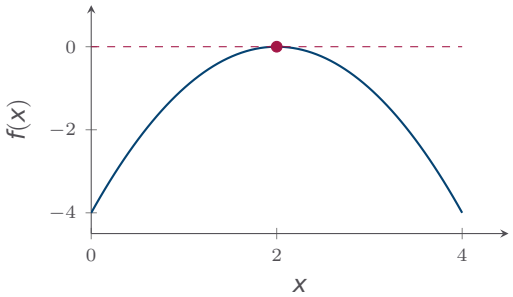
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The horizontal tangent is an upper bound on the entire graph.
Zero gradient means a flat tangent, not a constant function nearby.

Proof 2: return from the Lagrangian to utility

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The same slackness and feasibility minimise $\mathcal{L}(x^*, \lambda, \mu)$ over nonnegative multipliers: the saddle interpretation is recovered.

Budget exhaustion: Walras' law

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Assume an optimum exists. What would happen if it left strictly positive budget slack?

The key: with slack, every sufficiently small change in the consumption domain remains affordable.

Proof: LNS and budget exhaustion

Suppose an optimum x^* has slack $\delta = m - p \cdot x^* > 0$.

→ Back to budget exhaustion

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LNS supplies a strictly better such y in the consumption domain, contradicting optimality.

→ Back to budget exhaustion

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Convex preferences imply $z_t \succeq w$ for every such w. Thus $z_t \in D(p, m)$.

Hence $D(p, m)$ is convex. It can still be a whole line segment.

Strict convexity: at most one optimum

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Does strict convexity alone establish existence?

No. Continuity and compactness gave existence; strict convexity gives at most one optimum.

Proof: uniqueness from strict convexity

Suppose distinct $x, y \in D(p, m)$. Their midpoint z is feasible because the budget is convex.

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This contradicts x being optimal. There is at most one optimum; existence gives a unique one.

→ Back to uniqueness

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It is a function of the environment, not of a chosen bundle.

Indirect utility: the properties to prove

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- More wealth weakly increases v; under LNS the increase is strict.
- v is continuous on the positive-price, positive-wealth domain.

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Maximising the same utility over a smaller set cannot give a larger value: $v(q, m) \leq v(p, m)$.

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$p \cdot y \leq p \cdot x + \|p\|\|y - x\| < m + (m' - m)/2 < m'$.

So y is affordable at the new wealth and $v(p, m') \geq u(y) > v(p, m)$.

Continuity: put every budget on the same simplex

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For each budget, write $z_i = p_i x_i / m$, or equivalently $x_i = (m/p_i) z_i$.

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With $g(p, m, z) = u((m/p_1) z_1, \ldots, (m/p_L) z_L)$, we have $v(p, m) = \max_{z \in \Delta} g(p, m, z)$.

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No differentiability or uniqueness of the optimiser was needed.

A less obvious property: quasiconvexity

NES

For $0 \leq t \leq 1$, set $(p^t, m^t) = t(p, m) + (1 - t)(q, n)$.

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So the mixed budget cannot deliver utility above both endpoint values. → Proof

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→ Back to the property

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Thus every bundle affordable at the mixture is affordable at at least one endpoint.

Its utility cannot exceed $\max\{v(p, m), v(q, n)\}$. Maximise over the mixed budget.

→ Back to the property

Before differentiating, inspect the economics

Substitutes: compare utility gained per unit of spending; a corner may be optimal.

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Smooth interior problem: marginal conditions can identify a candidate.

Always check feasibility, boundary cases and a reason why the candidate is globally optimal.

Checkpoint

What established existence? What could establish uniqueness?

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What can go wrong if we check only the interior first-order equations?

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Now an interior method is justified; it was not assumed at the start.

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The log representation is concave on the positive domain.

The bundle does not change, but the numerical utility and multiplier generally do.

Solve the interior conditions

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$$x_1(p, m) = \frac{\alpha m}{p_1}, \qquad x_2(p, m) = \frac{(1-\alpha)m}{p_2}.$$

Interpret the solution

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If $\alpha = 1/2$, $p = (2, 1)$ and $m = 12$, demand is $(3, 6)$.

Which of these predictions is a modelling restriction rather than an accounting identity?

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Keep $\alpha = 1/2$ and $m = 12$. Raise p_1 from 2 to 3, holding $p_2 = 1$.

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Does an unchanged quantity of good 2 mean that its utility contribution is unchanged?

Example 2: perfect substitutes

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If the ratios are equal, every point on the budget line is optimal.

Verify a corner with the full KKT system

Let $u(x) = 2x_1 + x_2$, $p = (3, 1)$ and $m = 12$.

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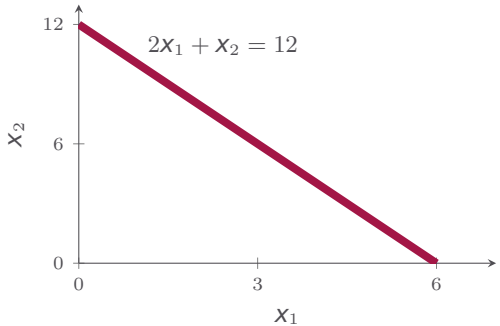
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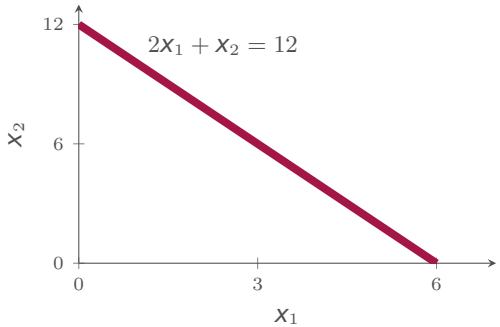
The zero quantity is optimal precisely because its marginal return is too low.

Equal returns: a whole set of demand



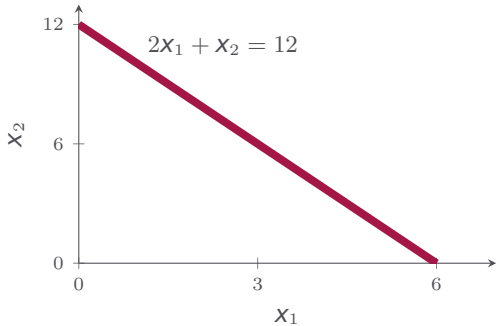
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Every budget-line bundle gives utility 12.
$D(p, m)$ is a segment, while $v(p, m) = 12$ is a single number.

Example 3: perfect complements

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At $p = (2, 1)$ and $m = 12$, the optimum is $(4, 4)$.

The optimum is a kink: the smooth stationarity formula does not apply there.

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If $m > 4$: $a^* > 0$, $\lambda = 1$, and extra wealth goes to a.

If $0 < m < 4$: $a^* = 0$, $\lambda = 2/\sqrt{m} > 1$, $\mu_a = \lambda - 1 > 0$.

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From a classroom solution to a structural model

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Fitting one choice does not establish the functional form. Use more variation and test the restrictions.

Compute indirect utility in the same example

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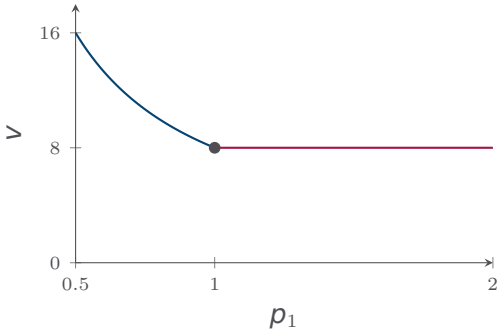
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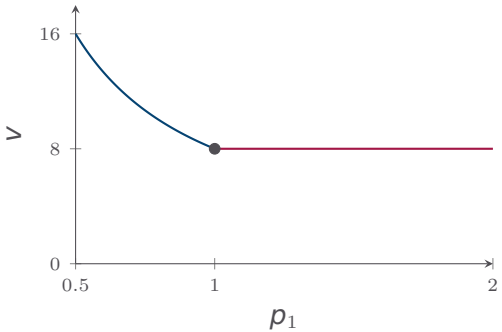
→ More than two goods

A continuous value need not be differentiable



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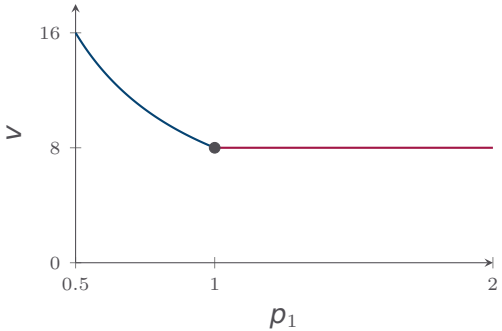
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At $p_1 = 1$, the consumer is indifferent along the budget line. The value has a kink.

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We need appropriate regularity and a differentiable value; a kink is a warning.

Differentiate the value, not the choice

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The wealth effect is positive when $\lambda^* > 0$; a higher price hurts in proportion to the quantity purchased.

The budget multiplier has an interpretation

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A binding budget does not by itself imply $v_m > 0$. → A counterexample

Roy’s identity: recover demand from value

Under the maintained consumer assumptions, suppose v is differentiable at (p, m) and $v_m(p, m) > 0$.

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Under these conditions the optimal quantity vector is unique. → Derivation

Check Roy in the Cobb–Douglas example

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$$x_1 = -\frac{-\alpha/p_1}{1/m} = \frac{\alpha m}{p_1}.$$

Likewise $x_2 = (1-\alpha)m/p_2$. The demand agrees with the direct solution.

Demand survives a change of utility scale

If $\widehat{v} = f(v)$ with $f'(v) > 0$, then

$$\widehat{v}_{p_i} = f'(v) v_{p_i}, \qquad \widehat{v}_m = f'(v) v_m.$$

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A strictly increasing transformation can have a zero derivative at a point; then the differentiable ratio may be unavailable.

Checkpoint

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What justifies treating a KKT candidate as globally optimal?

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Why can $D(p, m)$ be multivalued while $v(p, m)$ is single-valued?

What replaces equality of marginal returns for an unconsumed good?

What justifies treating a KKT candidate as globally optimal?

What two differentiability/positivity conditions are needed to use Roy's formula?

Read next, and keep the core route clear

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The next course block develops expenditure minimisation, Hicksian demand and duality. Reserve slides follow.

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Homework 1 covers Lectures 1–2 and is due on 14 November 2026.

Next block: two ways to ask the consumer question

Utility maximisation: given p, m, how much utility can the consumer attain?

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Utility maximisation: given p, m, how much utility can the consumer attain?

Expenditure minimisation: given $p, \bar{u}$, how little spending attains the target?

One fixes resources; the other fixes a level of the same preferences.

Expenditure minimisation

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e is the expenditure function; H is the Hicksian demand correspondence.

Assume the target is attainable. A singleton is denoted $h(p, \bar{u})$.

Why an unbounded feasible set can still have a minimum

Assume $p \gg 0$, continuous u, and pick a bundle $\widehat{x}$ with $u(\widehat{x}) \geq \bar{u}$.

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A continuous cost attains a minimum there; excluded bundles are more expensive. → Details

Hicksian demand: basic properties

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No excess utility uses continuity and positive prices, not LNS.

The expenditure function has a different curvature

Homogeneity of degree one: $e(tp, \bar{u}) = te(p, \bar{u})$.

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NES

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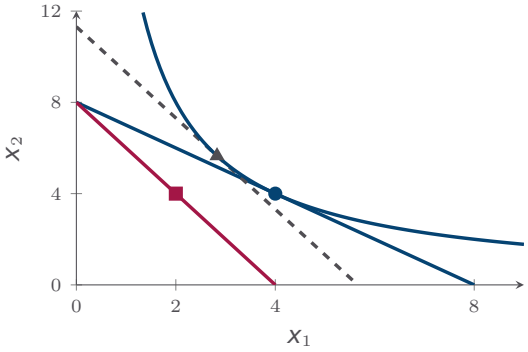
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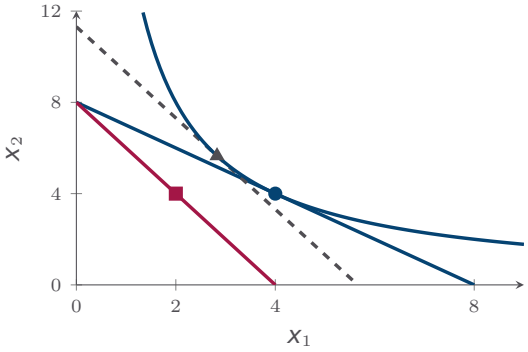
It is the infimum of linear cost functions. → Proofs

Fixed wealth versus fixed utility



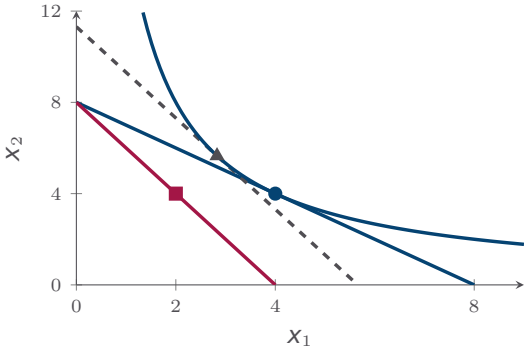
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$u = \sqrt{x_1 x_2}$; initially $p = (1, 1)$, $m = 8$, $x = (4, 4)$.
Raise p_1 to 2: Marshallian demand at the same wealth is $(2, 4)$.
To keep utility 4, spend $8\sqrt{2}$ and buy $(2\sqrt{2}, 4\sqrt{2})$.

Two identities to develop next time

NES

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For unique demands,

$$h\big(p, v(p,m)\big) = x(p,m).$$

These connect the two problems; they are not an assumption that the consumer literally solves both.

A monetary compensation question

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A money-metric change is useful, but summing it across people still involves welfare judgements.

Why the constraint qualification holds here

Active nonnegativity constraints have gradients e_i for the coordinates at zero.

→ Back to necessity and sufficiency

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The active gradients are therefore linearly independent. This includes corner optima.

→ Back to necessity and sufficiency

Proof: why concavity makes KKT sufficient

NES

For feasible y, concavity and differentiability imply

$$u(y) - u(x^*) \leq \nabla u(x^*) \cdot (y - x^*).$$

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Thus $u(y) \leq u(x^*)$ for every feasible y.

→ Back

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Both saddle inequalities hold. → Back to the bridge

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The function is strictly increasing and quasiconcave. That alone does not make stationarity sufficient.

→ Back

A binding budget with zero marginal value

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Roy's ratio becomes $0/0$ at that point, although demand $x = m/p$ is perfectly well-defined.

→ Back to the multiplier

Cobb–Douglas with any positive exponent sum

Let $u(x) = \prod_{i=1}^{L} x_i^{a_i}$ with $a_i > 0$ and $A = \sum_i a_i$.

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No restriction $A \leq 1$ is needed for this preference solution.

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If $v_m > 0$, every optimiser satisfies the same Roy formula, so demand is unique.

→ Back

Proof: expenditure minimisation has a solution

Choose an attainable target and a feasible $\widehat{x}$. Let

$$S = \{x \geq 0 : u(x) \geq \bar{u}\} \cap \{x : p \cdot x \leq p \cdot \widehat{x}\}.$$

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Cost is continuous, so it has a minimum on S. Every omitted feasible bundle costs more than $\widehat{x}$, so this is a global minimum.

→ Back to the preview

Proof: homogeneity and price concavity of expenditure

Scaling prices multiplies every feasible cost by the same positive number:

$$e(tp, \bar{u}) = te(p, \bar{u}), \qquad H(tp, \bar{u}) = H(p, \bar{u}).$$

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$$e(p^t, \bar{u}) = tp \cdot x^t + (1-t)q \cdot x^t \geq te(p, \bar{u}) + (1-t)e(q, \bar{u}).$$

→ Back to expenditure properties